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Q1) X company is capable of manufacturing three types of clothing: shirts, shorts, pants. The manufacture of each type of clothing requires that Mahendas have the appropriate type of machinery available

shirt machinery = $200 per week
short " = $150 " "
pants " = $100
} rent

(25)

Each week (two hours) of labor and (6000 sq yd) of cloth are available

Formulate an IP maximize weekly profit

type	labor-hours	cloth (sq yd)
shirt	3	4
short	2	3
pant	6	4

type	profit
shirt	6
short	4
pant	7

Q2) a) Build the network model of minimum spanning tree. (10 pts)

b) Using which one is more appropriate for solving small, medium and large sized MST problem instances

Prim's algorithm, Kruskal, or the model that you've built via a solver.

(20 pts)

"Questions"

Q1) x_1 = amount of shirts
x_2 = amount of shorts produced weekly
x_3 = amount of pants

$y_1 = \begin{cases} 1, & \text{if produced shirt} \\ 0, & \text{else} \end{cases}$

$y_2 = \begin{cases} 1, & \text{if produced short} \\ 0, & \text{else} \end{cases}$

$y_3 = \begin{cases} 1, & \text{if produced pants} \\ 0, & \text{else} \end{cases}$

$$\text{Max: } Z = 6x_1 + 4x_2 + 7x_3 - (200y_1 + 150y_2 + 100y_3)$$

$$3x_1 + 2x_2 + 6x_3 \leq 150$$
$$4x_1 + 3x_2 + 4x_3 \leq 160$$

$x_1 \leq M \cdot y_1$ M is a very large number
$x_2 \leq M \cdot y_2$
$x_3 \leq M \cdot y_3$

$x_1, x_2, x_3 \in \mathbb{N}$
$y_1, y_2, y_3 \in \{1, 0\}$

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Eğer her makine 150 saat çalışacaksa ve yd kısıtlamaları <u>her makine için</u> geçerli ise

$3x_1 \leq 150$ $\quad$ $4x_1 \leq 160$
$2x_2 \leq 150$ $\quad$ $7x_2 \leq 160$
$6x_3 \leq 150$ $\quad$ $4x_3 \leq 160$

olacaktı constraintler. Sonuç gerçeğe anlamsız/yazmasından diye ekstra not eklemek istedim.

Page 2

Q2)

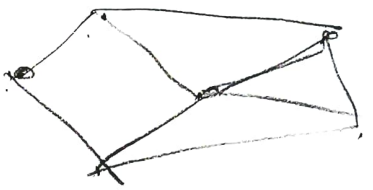
$$\min \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} \cdot d_{ij}$$

d_{ij} → cost between i and j

s.t

$$\sum_{i=1}^{n} a_{ij} = 1$$

$$\sum_{j=1}^{n} a_{ji} = 1$$



Solution is not a tour.

b) Kruskal would be more appropriate for small sized MST problem.

Prim would be more appropriate for medium sized MST problem.

Solver would be more appropriate for large sized MST problem.

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